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# Estimated municipality-level electric vehicle and passenger vehicle fleet composition across 14 European countries, 2011–2024

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We present a longitudinal dataset of passenger vehicle fleet composition at the municipality level for 14 European countries over 2011–2024. The dataset covers 71,715 Local Administrative Units (LAU) and disaggregates vehicle fleets into four categories: battery electric vehicles (BEVs), plug-in hybrid electric vehicles (PHEVs), fuel-cell electric vehicles (FCEVs), and other vehicles. City-level stocks are generated by applying an XGBoost-based spatial prediction model to official national statistics, using municipality-level GDP per capita, population, and road-masked nighttime light intensity as spatial allocation features. National totals for each passenger vehicle type are sourced from official IEA statistics. Model validation against independently collected data using sub-national data from Eurostat and municipality-level data from Norway and Sweden confirms satisfactory spatial accuracy ( $R^2 = 0.82\text{--}0.98$ ). The dataset enables comparative analysis of EV adoption geography, fleet electrification rates, and spatial disparities across European urban systems.

## Background & Summary

The electrification of passenger transport is reshaping mobility systems across Europe<sup>1–3</sup>. Understanding where electric vehicles are deployed requires data at the municipal level<sup>4</sup>. This spatial scale is crucial because adoption decisions, energy infrastructure, and social inequalities all operate here<sup>5,6</sup>. However, no public dataset provides consistent and comparable vehicle fleet composition estimates at this resolution<sup>7</sup>. Specifically, we lack harmonized records that span multiple cities and cover the entire timeline of the European EV transition<sup>8</sup>.

National vehicle registration statistics, while accurate in aggregate, are compiled by each country's transport authority to different temporal schedules, passenger vehicle classifications, and spatial granularities<sup>9</sup>. County- and city-level data are available for some countries, such as German Landkreise<sup>10</sup>, but are absent or non-publicly accessible for others. Eurostat, the statistical office of the European Union, compiles harmonized regional NUTS-2 passenger vehicle statistics across member states, but does not report the municipality-level detail required to capture within-country adoption heterogeneity. Harmonized cross-country datasets at the municipality level do not exist for the 2011 to 2024 period. This timeframe spans from the nascent EV market, which saw 16,856 BEVs across 14 countries in 2011, to a meaningful fleet scale of 7.69 million BEVs in 2024 across the study countries<sup>11</sup>.

Bridging this gap requires distributing national statistics to municipality level using predictors available at fine spatial resolution, an approach known as statistical downscaling. Tree-based ensemble methods, including XGBoost<sup>12</sup>, the algorithm used here, have been applied to disaggregate census counts into gridded population estimates<sup>13</sup>, municipal energy consumption and emissions<sup>14</sup> and global soil moisture<sup>15</sup>. We adopt the same strategy, combining national vehicle statistics with municipality-level population, GDP, and road-masked nighttime light intensity as predictors.

This dataset addresses that gap. We developed a novel spatial downscaling approach. This method calculates road light intensity using urban nighttime data combined with road networks as masks<sup>16</sup>. It then uses this light

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intensity, along with city-level socioeconomic attributes such as population and GDP, to build the urban training features. Finally, using national-level panel data as the training set, we predict city-level vehicle numbers. This method distributes official national vehicle stock totals to 71,715 municipalities, including BEV, PHEV, and FCEV counts. City-level stocks are generated for each passenger vehicle category, year, and municipality, and proportionally rebalanced to match national totals exactly. We compared the generated data with external data. The validation results against sub-national data from Eurostat and city-level data from Sweden and Norway show that  $R^2$  values range between 0.820 and 0.984, demonstrating its accuracy<sup>17</sup>.

The resulting dataset documents a period of rapid and spatially uneven change. BEV fleet share across the 14 study countries grew from 0.04% in 2011 to 7.46% in 2024. Adoption is highly spatially concentrated: Norway leads with a 2024 national EV fleet share of 35.0%, while the highest municipality-level shares reach 53.2%. Meanwhile, fleet electrification in southern European municipalities remains in early stages, with Italy and Spain both below 2% national EV fleet share in 2024. These patterns are captured in consistent, cross-nationally comparable form for the first time at municipality resolution<sup>18,19</sup>.

The dataset is intended for researchers in transport geography, urban sustainability, energy systems, and climate policy, as well as for practitioners requiring spatially disaggregated vehicle fleet baselines<sup>20–22</sup>. All code used to generate the dataset is made publicly available.

## Methods

An overview of the methods used in this study is provided in Fig. 1.

**Study area and period.** The dataset covers 14 European countries, Austria, Belgium, Switzerland, Germany, Denmark, Spain, Finland, France, United Kingdom, Italy, Netherlands, Norway, Portugal, and Sweden, over the period 2011–2024, 14 years. These countries collectively represent the principal EV markets in Europe and encompass a wide range of adoption trajectories, from early movers with high EV shares (Norway, the Netherlands, Sweden) to markets where the transition is still at early stages (Portugal, Spain, Italy)<sup>23,24</sup>. To determine the spatial boundaries of each city, we used Eurostat's Local Administrative Units (LAU) 2020 dataset, which provides a single, official municipal-level administrative division per country, pre-filtered to the 14 study countries (71,845 units).

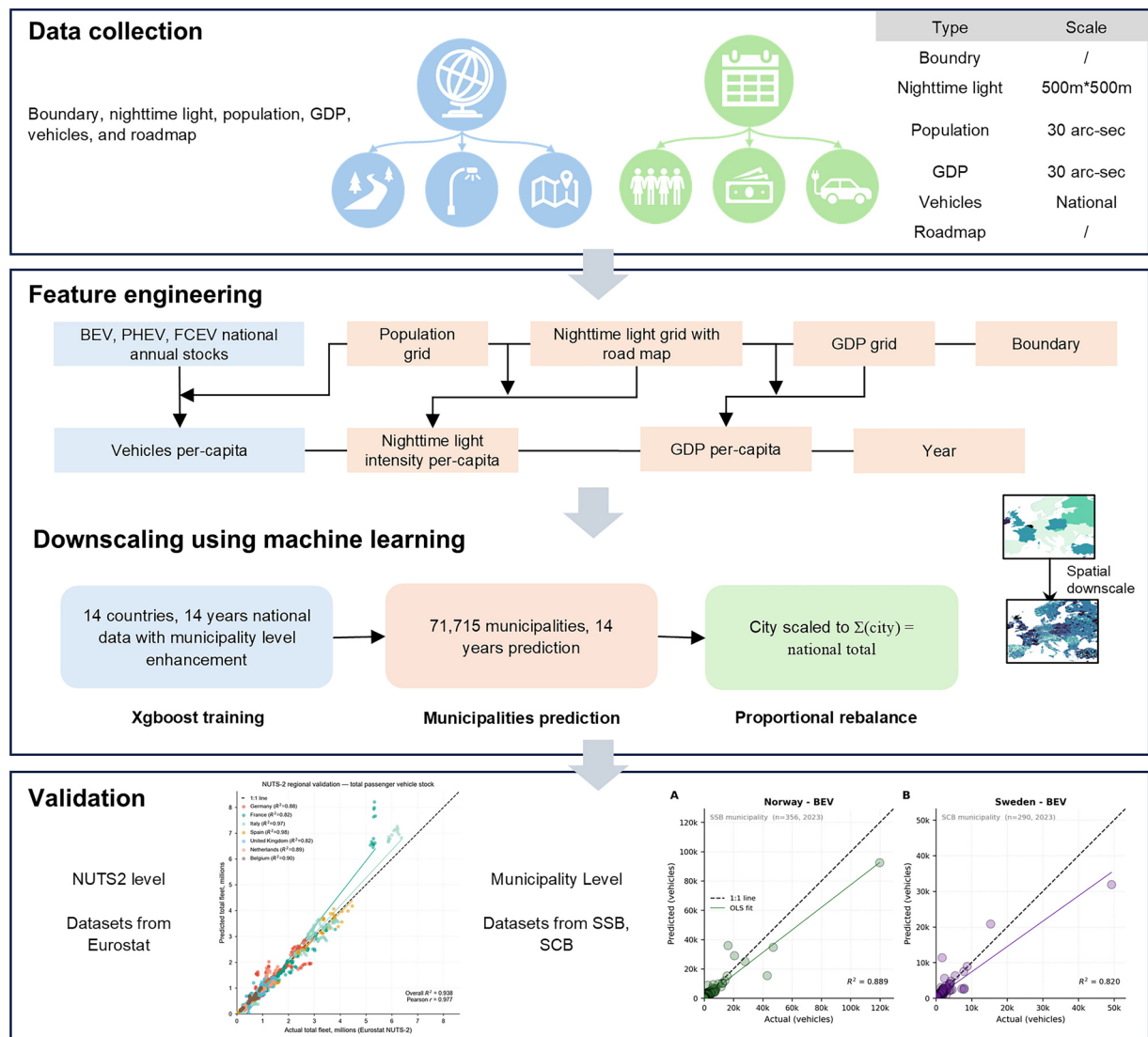
**Data sources.** *National vehicle stock.* Annual national vehicle stock by passenger vehicle type, BEV, PHEV, FCEV, and total fleet, were compiled by the International Energy Agency (IEA) Global EV Outlook 2026 dataset<sup>25</sup>. To account for the remainder of the fleet, the other-vehicles category represents all non-electrified passenger vehicles, primarily internal combustion engine vehicles. Its count is calculated by subtracting the combined number of battery electric, plug-in hybrid electric, and fuel cell electric vehicles from the total vehicle fleet. This total fleet size is itself derived by dividing the national count of electric vehicles by their corresponding share of the overall fleet. This approach reflects the structure of the IEA source data, which directly reports EV counts and EV fleet share rather than non-EV vehicle totals.

*Validation data.* Sub-national vehicle stock data used exclusively for model validation, were assembled from three independent sources. NUTS-2 regional passenger car stock was obtained from Eurostat, which reports total registered passenger cars by NUTS-2 region and year<sup>26</sup>. Data were extracted for seven countries with more than 10 NUTS-2 regions: Germany, France, Italy, Spain, the United Kingdom, the Netherlands, and Belgium. Municipality-level BEV stocks for Norway were obtained from Statistics Norway (SSB, Statistisk sentralbyrå), table 07849, which reports registered battery electric passenger cars by municipality<sup>27</sup>. The reference year is 2023, covering 356 municipalities. Municipality-level BEV and PHEV stocks for Sweden were obtained from Statistics Sweden (SCB, Statistiska centralbyrå)<sup>28</sup>, from the vehicle register tabulations by municipality. Annual files covering 2011–2023 were used, with the 2023 snapshot (290 municipalities) serving as the primary validation reference.

*Nighttime light intensity with road-mask.* City-level nighttime light statistics were derived from annual global NPP-VIIRS-like nighttime light data by Chen *et al.*<sup>29</sup>, clipped to road-masked zones within each municipality. This dataset reconstructs a NPP-VIIRS-like product for 1992–2024 using deep-learning at 500 m resolution, and annual composites of this dataset were constructed by taking the median of valid monthly observations for each pixel. Two metrics were extracted per city per year: light sum (total luminosity within the road-masked zone, a proxy for economic activity scale) and light mean (mean pixel intensity within the road-masked zone, a proxy for development density).

Road-masked zones were constructed by OpenStreetMap road network geometries onto the NTL raster grid as a binary mask and retaining only pixels that intersect a road segment<sup>30</sup>. Selected road categories include motorway, trunk, primary, secondary, tertiary, residential, and their respective links. The effective spatial unit of the mask is determined by the native NTL pixel resolution of approximately 500 m. Zonal statistics (sum and mean of road-masked NTL values) were then computed per municipality using the boundary polygons.

*GDP and population.* City-level GDP (PPP) and population data are sourced from Kummur *et al.*<sup>31</sup> and NASA GPW v4<sup>32</sup>, respectively at five-year intervals. Values for each municipality were obtained by summing all gridded pixels whose centroids fall within the boundary polygon. Annual interpolation was performed per municipality using piecewise linear interpolation. For years beyond the last available reference year (i.e., 2021–2024), values were linearly extrapolated using the slope of the final interval. The resulting annual series were clipped to zero to eliminate artefactual negative values that can arise from extrapolation in declining series. In practice, negative values occurred in fewer than 0.3% of city-year observations. Features derived from these variables



**Fig. 1** Research framework. Data collection assembles municipality boundaries, road-masked nighttime light (NTL) intensity (500 m resolution), population and GDP (30 arc-second gridded resolution), the OpenStreetMap road network used to build the road mask, and national-level BEV/PHEV/FCEV vehicle stock statistics. Feature engineering derives the per-capita predictors used by the model, vehicles per capita, road-masked NTL intensity per capita, GDP per capita, and year, from these inputs. The prediction stage trains a separate XGBoost regressor per vehicle type on the 14 country, 14-year national panel (augmented with Germany's Kreis-level registration data), generates predictions for all 71,715 municipalities, and proportionally rebalances city-level values so that each country's summed municipal total matches its official national total exactly. Validation compares model output against independent ground truth at NUTS-2 regional level (Eurostat) and at the municipality level (SSB, Statistics Norway; SCB, Statistics Sweden).

include GDP per capita, light sum per capita, and mean light per capita. Where a municipality's interpolated population is zero, all per-capita features are set to zero to avoid division-by-zero artefacts.

**Spatial prediction model.** A separate XGBoost regressor is trained for each passenger vehicle type (BEV, PHEV, FCEV, other)<sup>12</sup>. The training target is vehicles per capita. The core training set uses all available country-year observations (14 countries  $\times$  14 years = 196 observations per model). For BEV, PHEV, and other vehicles, this panel is augmented with ~1,197 additional cross-sectional observations built from Germany's Kreise (districts), using battery-electric, plug-in-hybrid, and total passenger car registration counts from the Kraftfahrt-Bundesamt (KBA) vehicle registration database (FZ 28), Landkreis (NUTS-3) level<sup>33</sup>, October 2023 snapshot (399 districts, the most recent publicly available disaggregated release at time of analysis): this was found to materially improve fine-grained spatial accuracy relative to training on the national panel alone, without degrading accuracy in the two countries (Norway, Sweden) whose own ground truth is not used in training.

| Parameter        | Value |
|------------------|-------|
| n_estimators     | 1000  |
| learning_rate    | 0.01  |
| max_depth        | 6     |
| subsample        | 0.8   |
| colsample_bytree | 0.8   |
| min_child_weight | 1     |
| reg_alpha        | 0.1   |
| reg_lambda       | 1.0   |

**Table 1.** Hyperparameters.

Because this KBA data enters model training directly, it is not also used as an independent validation source (see Technical Validation). FCEV is trained on the national panel only, since the KBA source has no FCEV breakdown.

The hyperparameter configurations are shown in Table 1. n\_estimators controls the number of boosting trees; learning\_rate controls the shrinkage applied to each tree's contribution; max\_depth controls the maximum tree depth (model complexity); subsample and colsample\_bytree control row/column subsampling per tree for regularisation; reg\_alpha/reg\_lambda control L1/L2 regularisation strength.

City-level counts are then proportionally rebalanced to match the official national totals exactly. Let  $V'_{i,c,t}$  denote the raw model prediction for municipality  $i$  in country  $c$  in year  $t$ , and let  $N_{c,t}$  denote the official national total for the same country, year, and passenger vehicle type. The rebalanced estimate is:

$$V_{i,c,t} = V'_{i,c,t} \times \frac{N_{c,t}}{\sum_{j \in c} V'_{j,c,t}}$$

where the sum in the denominator runs over all municipalities  $j$  belonging to country  $c$ . In the degenerate case where all city predictions for a given country-year-passenger vehicle combination are zero (which can occur for FCEV in early years), the rebalanced values are set to zero rather than applying an undefined ratio. This rebalancing step is applied independently for each country, year, and passenger vehicle type, ensuring the sum of city-level estimates equals the national total.

**Data cleaning and quality control.** *Alignment between boundaries and gridded socioeconomic data.* Municipality boundary polygons and the gridded GDP and population datasets are produced by independent sources and do not share a common spatial framework. Misalignment arises in two situations: (1) very small municipalities whose polygon area is smaller than a single grid cell, and (2) municipalities located at national borders where grid cells may be partially attributed to a neighboring country. In both cases, the zonal extraction may return no overlapping pixels, yielding a missing value for that municipality-year. This affects a small minority of municipalities: in a representative snapshot year (2020), 70 of 71,715 municipalities (0.10%) have a missing GDP value and 42 of 71,715 (0.06%) have a missing population value, for a total of 79 municipalities (0.11%) once both variables are combined. Missing values are treated as zero before interpolation, on the grounds that near-zero socioeconomic activity is the most defensible assumption for municipalities that fall below the grid resolution. The resulting zero-valued observations are retained in the dataset and receive zero-valued per-capita features, which the XGBoost model handles consistently with the zero-feature observations it was trained on.

*Overseas-territory exclusion.* French LAU units in Guadeloupe, Martinique, Guyane, Réunion, and Mayotte (130 of 71,845) are excluded from the released dataset. These territories fall outside the geographic window used to read the nighttime-light raster for continental Europe, so their road-masked light, GDP, and population features cannot be reliably extracted.

**Baseline model and feature attribution.** *Baseline model.* A baseline XGBoost regressor was trained for each vehicle type using only GDP per capita and year as predictors, omitting the nighttime light features. The baseline model otherwise followed the same training procedure, hyperparameter configuration (Table 1), and national-total rebalancing step as the full model, isolating the contribution of nighttime light to predictive accuracy.

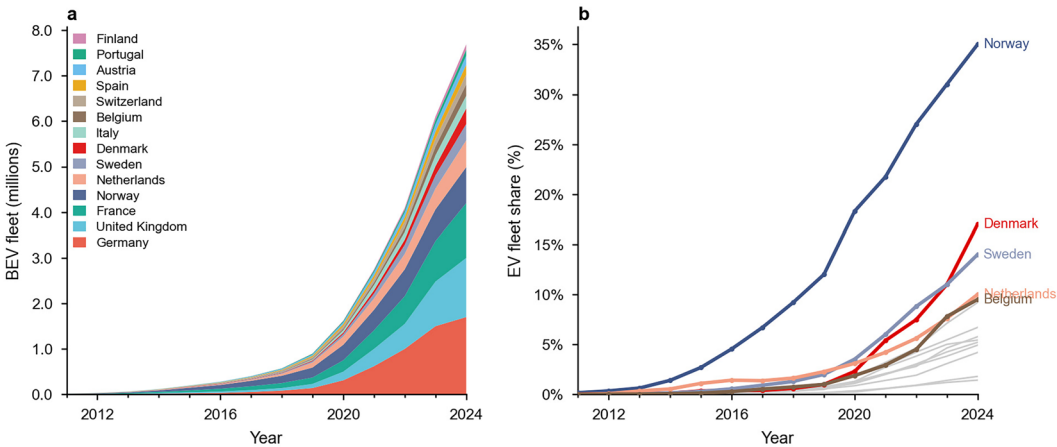
*Feature attribution.* SHAP (SHapley Additive exPlanations) values were computed for the full model using TreeExplainer[REF], separately for each vehicle type. SHAP values satisfy exact local additivity, so population-weighted averaging of absolute SHAP values across municipalities preserves this identity and yields an unbiased decomposition of feature importance. Values were computed at municipality level for the seven NUTS-2 validation countries and aggregated as population-weighted means, expressed as each feature's percentage contribution to the total attribution for each vehicle type.

## Data Records

All data are deposited at [<https://doi.org/10.6084/m9.figshare.32241864>]<sup>34</sup> and are described below (Table 2):

| Dimensions: 1,004,010 rows × 14 columns (71,715 municipalities × 14 years) |         |          |                                              |
|----------------------------------------------------------------------------|---------|----------|----------------------------------------------|
| Column                                                                     | Type    | Unit     | Description                                  |
| gisco_id                                                                   | string  | —        | Eurostat's globally-unique LAU identifier    |
| muni_name                                                                  | string  | —        | Municipality name                            |
| cntr_code                                                                  | string  | —        | Eurostat's native 2-letter country code      |
| country                                                                    | string  | —        | Country name                                 |
| year                                                                       | integer | —        | Calendar year (2011–2024)                    |
| BEV                                                                        | float   | vehicles | Estimated BEV stock                          |
| PHEV                                                                       | float   | vehicles | Estimated PHEV stock                         |
| FCEV                                                                       | float   | vehicles | Estimated FCEV stock                         |
| other_veh                                                                  | float   | vehicles | Estimated other vehicle stock                |
| pop                                                                        | float   | persons  | Municipality population                      |
| gdp                                                                        | float   | PPP      | Municipality GDP (PPP)                       |
| light sum                                                                  | float   | DN       | Nighttime light sum within road buffer       |
| light mean                                                                 | float   | DN       | Nighttime light mean within road buffer      |
| ev_share                                                                   | float   | fraction | EV fleet share (BEV+PHEV+FCEV) / total fleet |

Table 2. City-level panel dataset.

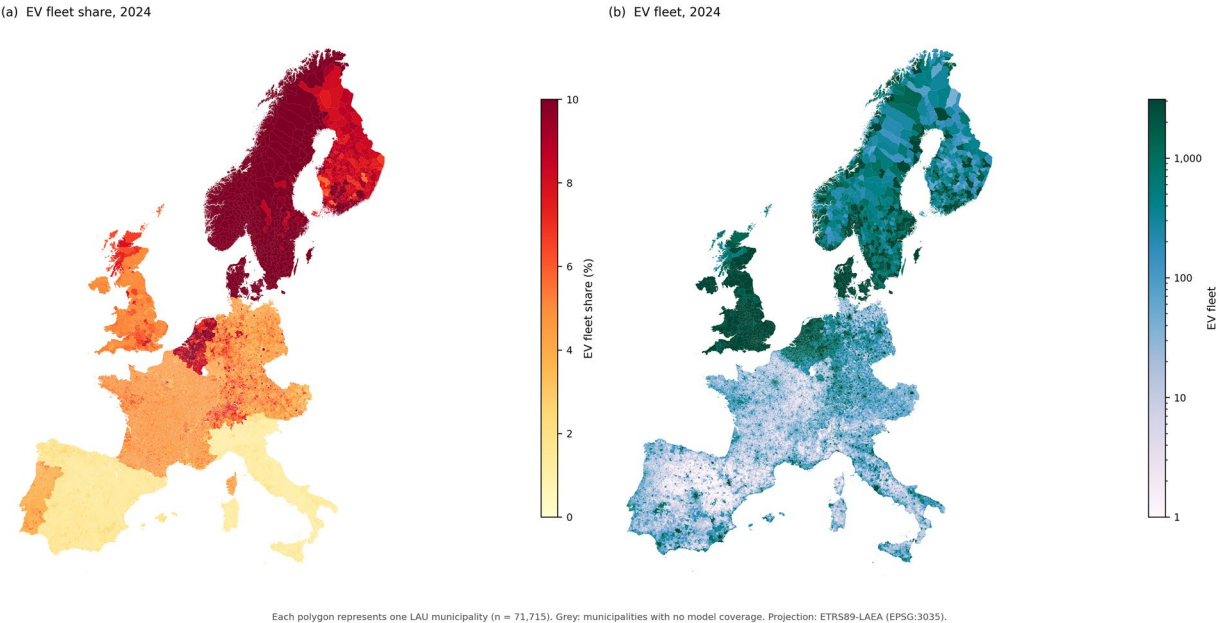


**Fig. 2** Temporal trends in national BEV fleet size and EV fleet share across 14 European countries, 2011–2024. Data source: IEA Global EV Outlook<sup>25</sup>. **(a)** Stacked area chart of absolute BEV fleet (millions of vehicles) by country, ordered by 2024 fleet size (largest at bottom). **(b)** EV fleet share (%) by country. The five countries with the highest 2024 fleet share, Norway (35.0%), Denmark (17.0%), Sweden (14.0%), Netherlands (10.0%), and Belgium (9.5%), are highlighted with colored lines and direct labels; the remaining nine countries are shown in grey.

**Data overview.** The dataset contains no missing values for vehicle stock columns. Vehicle stock values for very small municipalities may be fractional owing to per-capita scaling from national totals; these should be interpreted as expected values rather than integer counts.

Figure 2 shows the temporal trends of electric vehicles. The aggregate BEV fleet across the 14 study countries expanded from 16,856 vehicles in 2011 to 7.69 million in 2024, a 456-fold increase (Fig. 2a). Germany alone accounted for 22.1% of the 2024 total (1.70 million), followed by the United Kingdom (16.9%) and France (15.6%), reflecting the dominant influence of large-market absolute fleet size. Growth accelerated markedly after 2018 across all countries. However, Fig. 2b, which normalizes fleet size by national stock, reveals a starkly different picture of adoption intensity. Norway stands as an unambiguous outlier, with EV fleet share rising from 0.17% in 2011 to 35.0% in 2024. Denmark reached 17.0% and Sweden 14.0% by 2024, the Netherlands 10.0%, and Belgium 9.5%; the remaining nine countries all remained below 10%. The contrast between Germany’s large absolute fleet (1.70 M vehicles) and its modest 5.4% share, versus Norway’s smaller absolute fleet (790,000 vehicles) at 35.0%, illustrates that absolute market size and adoption intensity are largely decoupled.

Figure 3 maps the EV fleet share in panel a and the absolute EV fleet size in panel b across 71,715 municipalities with model coverage in 2024. A pronounced north-to-south gradient is the dominant spatial feature. Norwegian municipalities record a median EV fleet share of 31.1%, reaching a maximum of 53.2% in the highest adoption municipality. Denmark, Sweden, and the Netherlands form a second tier of early-adopting countries, with median shares of 17.0%, 12.1%, and 9.3%, and corresponding peaks of 26.3%, 23.8%, and 23.8%. At the opposite extreme, Spanish and Italian municipalities typically sit around a 1–1.5% median, with maximum adoption rates of 3.1% and 2.9% respectively. The four Nordic countries (Norway, Denmark, Sweden, Finland)



**Fig. 3** Geographic distribution of EV fleet share and EV fleet size across European municipalities, 2024. **(a)** EV fleet share (%) at municipality level. Color scale runs from 0 to 10% (99th percentile of the distribution). **(b)** Absolute EV fleet size per municipality on a logarithmic scale.

| Country        | NUTS-2 regions | Years covered | R <sup>2</sup> | Weighted MAPE | P50   |
|----------------|----------------|---------------|----------------|---------------|-------|
| Spain          | 19             | 2011–2020     | 0.984          | 8.2%          | 8.5%  |
| Italy          | 21             | 2011–2024     | 0.981          | 7.5%          | 7.1%  |
| Germany        | 38             | 2011–2024     | 0.951          | 6.5%          | 5.4%  |
| Netherlands    | 12             | 2011–2024     | 0.928          | 11.1%         | 6.5%  |
| United Kingdom | 36             | 2011–2018     | 0.906          | 10.0%         | 8.0%  |
| Belgium        | 11             | 2011–2024     | 0.906          | 7.8%          | 3.6%  |
| France         | 22             | 2011–2024     | 0.872          | 18.3%         | 19.6% |
| Overall        |                |               | 0.962          | 9.3%          | 7.2%  |

**Table 3.** NUTS-2 validation summary.

collectively hold 19.5% of the 2024 European EV fleet, while Spain, Italy, and Portugal combined account for 10.2%.

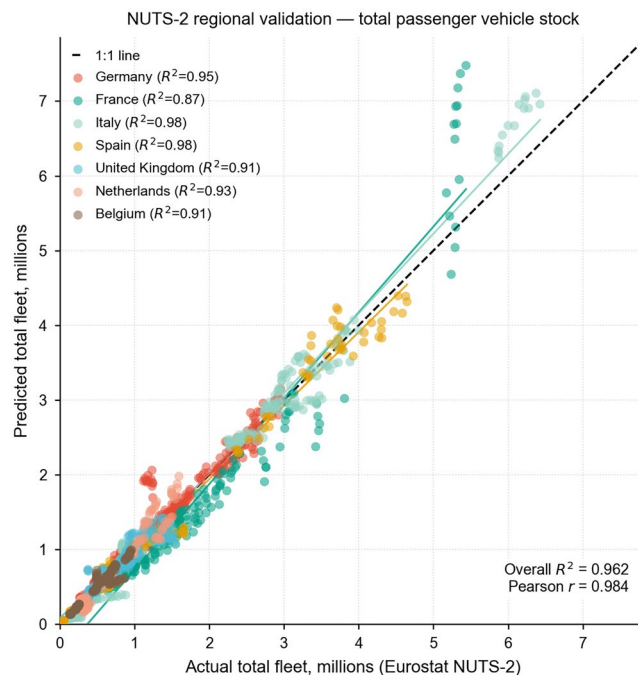
Technical Validation

The prediction model was validated at two spatial scales: NUTS-2 regional level using Eurostat administrative vehicle data across seven countries, and municipality level using independently collected sub-national ground-truth data from Norway and Sweden.

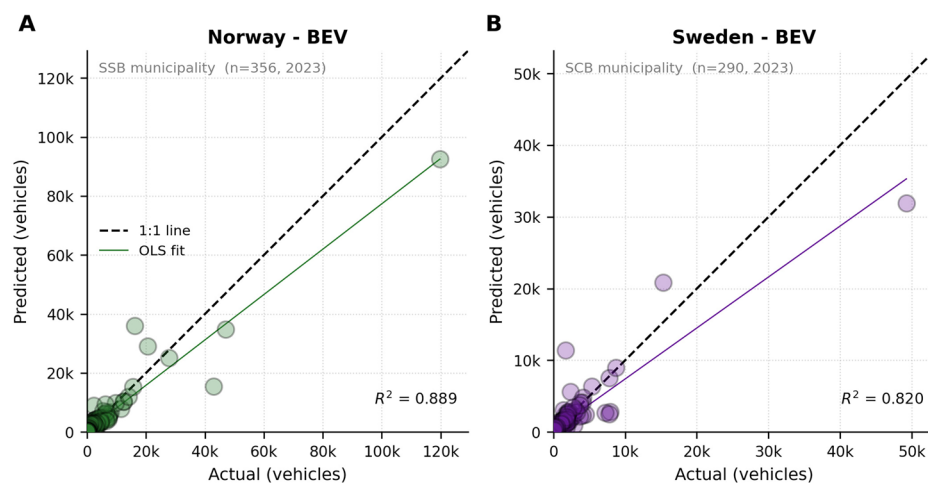
**NUTS-2 regional validation.** We validated the total number of NUTS-2 passenger vehicles in countries with more than 10 NUTS-2 regions, Germany (38), France (22), Italy (21), Spain (19), United Kingdom (36), Netherlands (12), and Belgium (11). Reference years vary by country depending on Eurostat data availability. Results are summarized in Table 3.

Figure 4 presents the NUTS-2 region validation results, plotting the model outputs against total passenger car ownership data reported by Eurostat for 159 NUTS-2 regions across seven countries, totaling 1,757 region-year observations. The overall fit is strong; a pooled R<sup>2</sup> of 0.962 (weighted MAPE 9.3%) indicates that the prediction model reproduces the cross-regional distribution of total fleet sizes at the NUTS-2 scale with high accuracy. Country-specific results show R<sup>2</sup> values exceeding 0.8 for all nations, demonstrating high model performance. Spain with an R<sup>2</sup> of 0.984 and Italy, at 0.981, show the highest precision. Germany at 0.951, the Netherlands at 0.928, the United Kingdom at 0.906, Belgium at 0.906, and France at 0.872 exhibit intermediate performance.

**Municipality-level validation.** BEV stocks were validated against independently collected ground-truth data for Norway and Sweden in 2023. As shown in Fig. 5 and Table 4, R<sup>2</sup> is 0.889 for Norway and 0.820 for Sweden, indicating that the spatial allocation pattern is well reproduced at municipality level for both genuinely independent out-of-sample checks. Vehicle-weighted MAPE is 34.0% for Norway and 36.1% for Sweden; the



**Fig. 4** NUTS-2 regional validation of modelled total passenger vehicle stock against Eurostat data. NUTS-2 regional validation of modelled total passenger vehicle stock against Eurostat data, full model. Each point represents one NUTS-2 region in one year. Colors denote countries; solid lines show per-country OLS regression fits; the dashed line is the 1:1 reference. Year coverage varies by country: United Kingdom 2011–2018, Spain 2011–2020, all others 2011–2024.



**Fig. 5** Municipality-level validation of modelled BEV fleet stocks against independent ground-truth data, 2023. Each panel displays a scatter plot of predicted versus actual vehicle counts, both axes on a compressed (k/M) count scale. A dashed black line represents the 1:1 relationship; a solid coloured line indicates the OLS regression fit. **(A)** Norway BEV against SSB municipality data ( $n=356$ ). **(B)** Sweden BEV against SCB municipality data ( $n=290$ ). Both panels use the reference year 2023, the most recent year with complete ground-truth coverage at time of analysis.

P50 values (53.4% and 34.1%) indicate that more than half of Norwegian municipalities individually show larger relative error than the weighted average, consistent with Norway's larger number of very small municipalities. In addition to  $R^2$ , MAE and RMSE were computed for both validation datasets: Norway BEV, MAE=659 and RMSE=2,539 vehicles; Sweden BEV, MAE=363 and RMSE=1,370 vehicles. The national model-to-reference ratios are 1.001 for Norway and 0.996 for Sweden, confirming that the proportional rebalancing procedure keeps systematic bias small at national scale. To summarize validation performance across both spatial scales in a single view, all nine  $R^2$  values, seven from the NUTS-2 regional validation and two from the municipality-level

| Type              | Norway       | Sweden       |
|-------------------|--------------|--------------|
| Data source       | SSB          | SCB          |
| Spatial unit      | Municipality | Municipality |
| n                 | 356          | 290          |
| Passenger vehicle | BEV          | BEV          |
| R <sup>2</sup>    | 0.889        | 0.820        |
| Weighted MAPE     | 34.0%        | 36.1%        |
| P50               | 53.4%        | 34.1%        |

**Table 4.** Municipality-level validation summary.

validation, are ranked and plotted together in Supplementary Fig. S1. Supplementary Fig. S4 shows the performance of representative Swedish municipalities of various sizes across the entire time series.

**Feature importance and feature ablation analysis.** To evaluate the incremental contribution of road-masked nighttime light, we trained a baseline model using only GDP per capita and year as predictors, applying the same prediction and rebalancing procedure as the full model. Pooling seven countries and 1,757 region-year observations at NUTS-2 level, the full model improved the baseline, reaching  $R^2=0.962$  and weighted MAPE=9.3% compared with  $R^2=0.953$  and weighted MAPE=10.6% for the baseline, shown in Supplementary Fig. S2(b). The largest gains from adding road-masked nighttime light occurred in Italy, with its  $R^2$  value rising from 0.959 to 0.981, a gain of 0.022, followed by Spain, with an increase of 0.018; Germany, France, Belgium, and the Netherlands showed little change or a small decrease in  $R^2$  despite consistently lower MAPE with road-masked nighttime light included.

We computed SHAP feature attributions on the full model, calculated at the municipality level and aggregated by population-weighted mean across the seven NUTS-2 validation countries. For the BEV model, the temporal trend feature (year) accounts for 48.5% of the total attribution, GDP per capita 24.6%, and the two road-masked nighttime-light features jointly 26.9%, comprising 16.8% from road-masked light sum per capita and 10.1% from road-masked mean light per capita, shown in Supplementary Fig. S2(a). For other vehicle, road-masked nighttime light accounts for a larger share, 38.8%, exceeding GDP's contribution of 17.7%. These results indicate that road-masked nighttime light captures spatial variation not explained by GDP and population.

**Rebalancing adjustments.** To quantify the magnitude of the proportional rebalancing step, we calculated the total fleet rebalancing coefficient for each of the seven NUTS-2 validation countries (Supplementary Fig. S3). The rebalancing coefficients for these seven countries ranged from 0.86 to 1.22: specifically, the United Kingdom (0.86), Spain (0.89), France (0.93), Belgium (0.93), the Netherlands (0.96), Germany (1.08), and Italy (1.22).

### Usage Notes

The city-level panel (Table 2) is the primary analysis-ready file for most research applications. Municipality boundary geometries are not included in the released file, users who need to map the data should join it to their own copy of the Eurostat LAU 2020 boundaries directly via the `gisco_id` column, which is Eurostat's own globally-unique identifier for each municipality.

Users should note that municipality-level values in this dataset are model estimates, not observed registration data. Accuracy has room for improvement, particularly for small municipalities and rare powertrain categories with low counts; future work will incorporate additional predictor variables to improve model accuracy. EV adoption is also influenced by factors not included in our model, such as charging infrastructure availability, housing characteristics such as access to off-street parking, and local policy incentives such as company-car tax treatment. These omitted variables likely contribute to the residual cross-country variation in validation performance reported above. We searched for additional municipality-level vehicle registration data, particularly for lower-EV-adoption countries such as Spain, Italy, and Portugal, but were unable to identify publicly available municipality validation data at comparable granularity. Our NUTS-2 regional validation already includes Spain and Italy at a coarser spatial resolution.

LAU units vary substantially in population and area across countries. France, for example, has approximately 34,838 communes, far more fine-grained than the other study countries. This heterogeneity means municipality-level statistical noise varies by country; we recommend that cross-country comparisons of absolute municipality-level values be interpreted cautiously, or performed after aggregating to a more comparable spatial scale such as NUTS-3 or NUTS-2. We employed linear extrapolation for GDP and population data for the 2021–2024 period; while this maintained consistency with the interpolation, it did not account for factors such as COVID-19. Future research could build upon this framework by substituting these with the latest gridded data.

### Data availability

The dataset is deposited at [<https://doi.org/10.6084/m9.figshare.32241864>]<sup>34</sup>. All input data used are publicly available: IEA Global EV Outlook statistics<sup>25</sup>; Eurostat regional transport data<sup>26</sup>, which is open data; Eurostat LAU administrative boundaries, openly published, with geometries not redistributed and only municipality identifiers referenced<sup>35</sup>; Kumm *et al.*'s downscaled GDP dataset, openly published<sup>31</sup>; NASA GPW v4 population data<sup>32</sup>, openly published; the Chen *et al.* nighttime-light product<sup>29</sup>, openly published; OpenStreetMap road network data

(ODbL)<sup>30</sup>, openly published; and the KBA<sup>33</sup>, SSB<sup>27</sup>, and SCB/Trafikanalys<sup>28</sup> municipality-level vehicle registration data used for training augmentation and validation, all publicly available from the respective national statistical agencies.

## Code availability

All code used to generate the dataset is available at [<https://github.com/UrbanModeling/eu-ev-municipality-fleet>].

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## References

1. Bibra, E. M. *et al.* Global EV outlook 2022: Securing supplies for an electric future (2022).
2. Victoria, M., Zeyen, E. & Brown, T. Speed of technological transformations required in Europe to achieve different climate goals. *Joule* **6**, 1066 <https://doi.org/10.1016/j.joule.2022.04.016> (2022). +.
3. Link, S., Schneider, L., Stephan, A., Weymann, L. & Plötz, P. Feasibility of meeting future battery demand via domestic cell production in Europe. *Nature Energy* **10**, 526–534 <https://doi.org/10.1038/s41560-025-01722-y> (2025).
4. Esmaili, A., Oshanreh, M. M., Naderian, S., MacKenzie, D. & Chen, C. Assessing the spatial distributions of public electric vehicle charging stations with emphasis on equity considerations in King County, Washington. *Sustainable Cities and Society* **107** <https://doi.org/10.1016/j.scs.2024.105409> (2024).
5. Guo, Z. H., You, L. L., Zhu, R., Zhang, Y. & Yuen, C. A City-scale and Harmonized Dataset for Global Electric Vehicle Charging Demand Analysis. *Scientific Data* **12** <https://doi.org/10.1038/s41597-025-05584-7> (2025).
6. Mai, L. *et al.* A high-resolution dataset on electric passenger vehicle characteristics in China and the European Union. *Scientific Data* **12**, 1449 <https://doi.org/10.1038/s41597-025-05770-7> (2025).
7. Pamidimukkala, A., Kermanshachi, S., Rosenberger, J. M. & Hladik, G. Barriers and motivators to the adoption of electric vehicles: A global review. *Green Energy and Intelligent Transportation* **3** <https://doi.org/10.1016/j.geits.2024.100153> (2024).
8. Onat, N. C. & Kucukvar, M. A systematic review on sustainability assessment of electric vehicles: Knowledge gaps and future perspectives. *Environmental Impact Assessment Review* **97** <https://doi.org/10.1016/j.eiar.2022.106867> (2022).
9. Lekaki, D. *et al.* Road transport emissions in EDGAR (Emissions Database for Global Atmospheric Research). *Atmospheric Environment* **324** <https://doi.org/10.1016/j.atmosenv.2024.120422> (2024).
10. Kraftfahrt-Bundesamt. Verkehr in Kilometern der deutschen Kraftfahrzeuge im Jahr 2016 (2016).
11. Cavalcante, I. *et al.* Dataset on Electric Road Mobility: Historical and Evolution Scenarios until 2050. *Scientific Data* **11** <https://doi.org/10.1038/s41597-024-03801-3> (2024).
12. Chen, T. & Guestrin, C. Xgboost: A scalable tree boosting system. *Proceedings of the 22nd acm sigkdd international conference on knowledge discovery and data mining*, 785–794 (2016).
13. Chen, Y. H., Xu, C. C., Ge, Y., Zhang, X. X. & Zhou, Y. N. A. 100 m gridded population dataset of China's seventh census using ensemble learning and big geospatial data. *Earth System Science Data* **16**, 3705–3718 <https://doi.org/10.5194/essd-16-3705-2024> (2024).
14. Patil, S., Pflugrad, N., Weinand, J. M., Kropp, J. & Stolten, D. Disaggregated Municipal Energy Consumption and Emissions in End-use Sectors in Germany and Spain for 2022. *Scientific Data* **12** <https://doi.org/10.1038/s41597-025-05938-1> (2025).
15. Zhang, Y. F. *et al.* Generation of global 1 km daily soil moisture product from 2000 to 2020 using ensemble learning. *Earth System Science Data* **15**, 2055–2079 <https://doi.org/10.5194/essd-15-2055-2023> (2023).
16. Chen, X. & Nordhaus, W. D. Using luminosity data as a proxy for economic statistics. *Proceedings of the National Academy of Sciences* **108**, 8589–8594 (2011).
17. Pocketbook, S. EU transport in figures. *Publications Office of the European Union, Luxembourg* (2016).
18. Li, G. L. *et al.* City-scale synthetic individual-level vehicle trip data. *Scientific Data* **10** <https://doi.org/10.1038/s41597-023-01997-4> (2023).
19. Imre, S., Çelebi, D. & Asan, U. Estimating potential adoption rate of electric vehicles in urban logistics. *Transportation Planning and Technology* **47**, 370–399 <https://doi.org/10.1080/03081060.2023.2287138> (2024).
20. Isik, M., Dodder, R. & Kaplan, P. O. Transportation emissions scenarios for New York City under different carbon intensities of electricity and electric vehicle adoption rates. *Nature Energy* **6**, 92–104 <https://doi.org/10.1038/s41560-020-00740-2> (2021).
21. Diao, M., Kong, H. & Zhao, J. H. Impacts of transportation network companies on urban mobility. *Nature Sustainability* **4**, 494–500 <https://doi.org/10.1038/s41893-020-00678-z> (2021).
22. Colak, S., Lima, A. & González, M. C. Understanding congested travel in urban areas. *Nature Communications* **7** <https://doi.org/10.1038/ncomms10793> (2016).
23. Mangipinto, A. *et al.* Impact of mass-scale deployment of electric vehicles and benefits of smart charging across all European countries. *Applied Energy* **312** <https://doi.org/10.1016/j.apenergy.2022.118676> (2022).
24. Kapousizis, G., Sarker, R., Ulak, M. B. & Geurs, K. User acceptance of smart e-bikes: What are the influential factors? A cross-country comparison of five European countries. *Transportation Research Part a-Policy and Practice* **185** <https://doi.org/10.1016/j.tra.2024.104106> (2024).
25. International Energy, A. Global EV Outlook 2026: Moving towards increased affordability. (International Energy Agency, Paris, 2026).
26. Eurostat. Regional transport. <<https://ec.europa.eu/eurostat/web/transport/information-data/regional-transport>>.
27. Statistics, N. Registered vehicles, by type of transport and type of fuel: Table 07849. <<https://www.ssb.no/en/statbank/table/07849>>.
28. Transport, A. Vehicle statistics. <<https://www.trafa.se/en/road-traffic/vehicle-statistics/>>.
29. Chen, Z., Liao, L., Wang, C., Shi, K. & Yu, B. The 1992–2024 Global NPP-VIIRS-like Nighttime Light Annual Data from Deep Learning Super-Resolution Reconstruction. *Journal of Remote Sensing* **6**, 0874 <https://doi.org/10.34133/remotesensing.0874>
30. OpenStreetMap, c. OpenStreetMap. <<https://www.openstreetmap.org>>.
31. Kumm, M., Kosonen, M. & Masoumzadeh Sayyar, S. Downscaled gridded global dataset for gross domestic product (GDP) per capita PPP over 1990–2022. *Scientific Data* **12**, 178 <https://doi.org/10.1038/s41597-025-04487-x> (2025).
32. Center for International Earth Science Information Network, C. U. Gridded Population of the World, Version 4 (GPWv4): Population Count. <<https://doi.org/10.7927/H4JW8BX5>> (2018).
33. Kraftfahrt, B. Fahrzeugarten (Vehicle Types Statistics). <[https://www.kba.de/DE/Statistik/Fahrzeuge/Fahrzeugarten/fahrzeugarten\\_node.html](https://www.kba.de/DE/Statistik/Fahrzeuge/Fahrzeugarten/fahrzeugarten_node.html)>.
34. Jin, Y. & Sharifi, A. Estimated municipality-level electric vehicle and passenger vehicle fleet composition across 14 European countries, 2011–2024. <https://doi.org/10.6084/m9.figshare.32241864> (2026).
35. Eurostat. (ed Gisco) (Eurostat, European Commission, Luxembourg, 2020).

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### Additional information

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